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AI-enabled vision systems for human-centered order picking – A design science research approach

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ABSTRACT

Digital technologies are critical in advancing a human-centered approach to warehouses that account for productivity and staff well-being. These technologies generate data addressing the negative conditions affecting the well-being of staff during order picking (OP), a labour intensive activity. This study analyzes artificial intelligence (AI)-enabled vision systems to enhance human-centricity and improve the generation and analysis of information about tasks executed by staff in OP. The study presents results from a pilot study in automotive manufacturing applying a design science research approach. The results show that AI-enabled vision systems enhance task identification, analysis, and efficiency in OP. The study suggests five actions including staff information, data acquisition, access restriction, data storage, and protection addressing the privacy concerns of these systems. The study discusses how these systems can integrate staff well-being by identifying human factors and outcomes. It offers three contributions: (1) an overview of activities for collecting task information through AI-enabled vision systems in human-centered OP; (2) evidence that existing architectures for human-centered manufacturing are essential for managing privacy implications; and (3) a discussion of the systems' impact on human factors and performance, and guidelines for developing and implementing these systems in future studies and operational environments.

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Artificial intelligence; machine learning; vision systems; warehousing 5.0; smart production logistics

1. Introduction

Industry 5.0 introduces a new paradigm that complements the current focus on digital technologies by emphasising human-centered, sustainable, and resilient manufacturing (Leng et al. 2023; Lu et al. 2022). The limited attention to socio-environmental demands in smart manufacturing, along with the shift from a technology-centric approach, drives the need for Industry 5.0 (Ghobakhloo et al. 2022; Ivanov 2023). Central to Industry 5.0 is its focus on human-centricity, which places worker well-being at the core of the manufacturing process, highlighting employment growth, workplace dignity, autonomy, and job satisfaction (Grosse et al. 2023; Rožanec et al. 2023; Slama et al. 2024).

A human-centered approach significantly impacts processes, tasks, infrastructure, competencies, and staffing requirements in warehouses (Grosse 2024; Ivanov 2023; Neumann et al. 2021; Winkelhaus, Grosse, and Morana 2021). This approach extends previous research that links tasks of order picking (OP) such as setup, travel, search, and picking to human factors (Grosse et al. 2015). It represents a shift in focus from optimising time, cost, and quality to addressing human factors such as workload, fatigue, motivation, injuries, communication, and supervision (De Lombaert et al. 2023; Falkenberg and Spinler 2023; Kumar, Narkhede, and Jain 2021).

Despite increasing interest in human-centered practices, both warehouse research and operations continue to fall short in achieving tangible improvements leaving staff under continued pressure (Brunetti, Gena, and Vernero 2022; De Lombaert et al. 2023; Lucchese and Mummolo 2024). Tasks in warehouses are labour intensive, and staff must manage an increasing diversity of products, faster deliveries, and smaller batch sizes (Ali and Phan 2022; Diefenbach, Grosse, and Glock 2024). The combination of these factors have negative consequences for the well-being of staff including increased stress and errors, poor ergonomics, and disengagement, which consequently reduce operational performance and increase costs (Cimini, Lagorio, Pirola, et al. 2021; Grosse 2024; Grosse et al. 2023; Mahroof 2019). These challenges are particularly evident in OP, which involves retrieving items from storage to fulfil orders, but includes

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a low level of automation and must address the needs of diverse staff (de Koster, Le-Duc, and Roodbergen 2007; De Lombaert et al. 2023).

The literature emphasises that data originating from digital technologies is essential to address the negative conditions affecting the well-being of staff in OP (Grosse 2024; Kumar, Narkhede, and Jain 2021; Winkelhaus, Grosse, and Morana 2021). These technologies enable real-time information flow, improving traceability, flexibility, and coordination in the movement of materials and information (Cimini, Lagorio, Romero, et al. 2021). Examples include Industrial Internet of Things (IIoT) devices, cloud based technologies, Artificial Intelligence (AI), big data analytics, and collaborative robots (Ciccarelli et al. 2022; Menti, Romero, and Jacobsen 2023; Rescio et al. 2023). For instance, Vijayakumar et al. (2022) proposed a framework integrating processes, system settings, and digital technologies to integrate human factors and operational performance in production logistics. Winkelhaus, Grosse, and Morana (2021) identified digital technologies that support the physical, mental, perceptual, and psychosocial tasks involved in OP. De Lombaert et al. (2023) discussed how incorporating human factors into OP improves operational performance and worker well-being. Glock et al. (2021) and Grosse et al. (2023) identified devices that account for human factors for staff in warehousing. Lucchese and Mummolo (2024) conducted laboratory experiments to assess the impact of digital technologies on cognitive tasks in OP, focussing on performance and staff well-being. Recent efforts to integrate digital technologies into OP have included IIoT devices, such as chest bands, wristbands, smart glasses, inertial measurement units, or company databases for monitoring staff tasks (Ciccarelli et al. 2022; Menti, Romero, and Jacobsen 2023; Pistolesi, Baldassini, and Lazzerini 2024; Rescio et al. 2023). However, staff often perceive these technologies as invasive, uncomfortable, or intrusive (Rescio et al. 2023). Additionally, signal quality can be subjected to external disturbances, and even after collecting information, its analysis remains predominantly manual, limiting the timely identification and resolution of negative working conditions (Kadir and Broberg 2020).

AI-enabled vision systems present an alternative complementing existing digital technologies for collecting data and supporting the advancement of a human-centered approach in OP. AI-enabled vision systems utilise network cameras, software, and communication technologies to analyze humans, machines, and products (Helm, Malikova, and Kembro 2023; Konstantinidis et al. 2023). AI enhances these systems by enabling the real-time acquisition of unstructured data, such as sequences of frames with complex visual and temporal information, and providing feedback to staff (Lou et al. 2022). These systems learn from historical data to classify images, enabling recognition or localisation (Smith, Smith, and Hansen 2021). They provide distinct advantages including the automatic collection of information, error recognition, the presentation of corrective actions, and analysis of the underlying causes of errors (Benmessabih et al. 2024; Generosi et al. 2022; Helm, Malikova, and Kembro 2023). Accordingly, these capabilities may reduce the cost of analysis and time required to identify and correct practices that negatively impact the well-being of staff (Battini et al. 2022; Shahin et al. 2024; Slama et al. 2024).

In the past, the implementation of AI-enabled vision systems in warehouses has yielded valuable insights. However, several key considerations related to advancing a human-centered approach in OP remain underexplored. Previous studies have employed these systems to capture motion data for ergonomic evaluations (Zaccaria et al. 2021), or to reduce musculoskeletal disorders while improving productivity (Battini, Persona, and Sgarbossa 2014; Diete et al. 2017; Zhao et al. 2022; Zheng et al. 2024). To extend these contributions, it is essential to collect data on specific OP activities–such as setup, travel, search, and picking–and to analyze the sequence of events that better consider staff well-being (Castaño-Amoros, Fuentes, and Gil 2023; Helm, Malikova, and Kembro 2023; Konstantinidis et al. 2023). At the same time, these applications must address growing privacy concerns associated with monitoring tasks and collecting behavioural data from employees (Brunetti, Gena, and Vernero 2022; De Lombaert, Rijal, et al. 2024; Mahroof 2019). This is particularly important for complying with regulations including the AI Act (European Parliament, Council of the European Union 2024), which establishes a legal framework for developing human-centric AIs. Accordingly, these systems must be designed to collect information without compromising workers' privacy, security, autonomy, or dignity, which can lead to resistance due to perceived excessive surveillance (Grosse et al. 2023).

The purpose of this study is to analyze the use of AI-enabled vision systems for enhancing the generation and analysis of information about tasks in OP of warehouses. The study applies a design science research (DSR) approach with a context–intervention–mechanism–outcome (CIMO) logic on a case in automotive manufacturing. The study focuses on a case in manufacturing warehouses, which offer a valuable context for examining a human-centered approach in OP. These environments rely on a large workforce and face simultaneous demands such as managing higher costs, supporting product customisation, ensuring operational flexibility,

and handling a broad range of components (de Koster, Le-Duc, and Roodbergen 2007; Kumar, Narkhede, and Jain 2021; Wang et al. 2025). The study presents three contributions to the literature. First, it describes the activities involved in generating and analyzing information when applying AI-enabled vision systems in human-centered OP. Second, it demonstrates how existing architectures that prioritise human-centered manufacturing can be applied to address privacy concerns related to the use of these systems. Third, it discusses the implications of generating and analyzing information when applying AI-enabled vision systems to human factors and staff performance. Finally, the study presents managerial implications, including guidelines for applying AI-enabled vision systems in OP.

2. Basic concepts

This section presents the basic concepts of human-centered OP, the architectures that support human-centricity, and AI-enabled vision systems. It describes current gaps in the literature and outlines how this study extends existing frameworks to address these gaps.

2.1. Human-centered order picking

A human-centered approach in OP is important because of the inherent involvement of staff and the labour intensive nature of its tasks (Vijayakumar et al. 2022). The literature on OP has experienced a change of priorities for achieving a human-centered approach (De Lombaert et al. 2023). In the past, the notion of a human-centered approach focussed exclusively on defining constraints of OP to prevent injuries or address its ergonomic aspects (Grosse et al. 2015). However, recent studies emphasise redefining its outcomes, considering human factors, and applying digital technologies to better consider staff well-being (Winkelhaus, Grosse, and Morana 2021). This expanded perspective has facilitated improvements in inclusivity (Lucchese, Panagou, and Sgarbossa 2024b; Ranasinghe et al. 2024), support for staff with disabilities (Rojas et al. 2024), movement efficiency and energy consumption (Lucchese, Digiesi, and Sgarbossa 2024), and the provision of cognitive assistance (Loske et al. 2024).

Previously, in a technology-centered approach, outcomes of OP prioritised operational performance and completion of tasks in a minimum time (de Koster, Le-Duc, and Roodbergen 2007). However, concerns over work rates, worker behaviour, execution, experience, and socio-demographic differences increasingly challenge this view (De Lombaert et al. 2023). Instead, a human-centered approach considers factors affecting the well-being of staff, such as procedural deviations, safety lapses, and staff fatigue (Glock et al. 2017; Leroy et al. 2024). Integrating this approach has lead to improvements in productivity, quality, and worker health supported by more accurate models of OP (Grosse et al. 2015; Vijayakumar et al. 2022). Productivity determines the time and cost required to complete an activity. Quality involves the rate of error during the activity. Well-being encompasses engagement, job satisfaction, safety, fatigue, workload, discomfort and injury risk (Sgarbossa et al. 2020; Winkelhaus, Grosse, and Glock 2022). Worker health addresses the physically demanding aspects and potential poor working conditions experienced by staff in OP (De Lombaert et al. 2023; Glock et al. 2019).

Human factors of OP, which define the interactions between staff and warehouse systems, are equally important in a human-centered approach (Pasparakis, Vries, and Koster 2023; Sgarbossa et al. 2020). The combination of these factors and the use of digital technologies are critical as their interaction determines the outcomes, operation, and consequences for staff (Grosse et al. 2023). Human factors including perceptual, cognitive, physical and psychological have an impact on the working conditions, capabilities, inter-dependencies and performance of staff (Grosse et al. 2015). Figure 1 presents the relation between human factors, tasks, and outcomes of OP according to Grosse et al. (2015). Perceptual factors involve detecting and perceiving necessary cues and information (Battini et al. 2015). Cognitive factors include processing information, affecting mental load, memory demands, and procedures for special cases or error recovery (Grosse et al. 2023). Physical factors include activities such as travelling between racks, carrying loads, pushing or pulling trolleys, stretching, bending, lifting, and extracting items (Grosse et al. 2015). Psychosocial factors include motivation, feedback, incentives, stress, boredom, work satisfaction, time pressure, and health of staff (Thylén, Wänström, and Hanson 2023).

Digital technologies significantly impact task execution, improving workplaces and increasing employee satisfaction and motivation in OP operations (Winkelhaus, Grosse, and Glock 2022; Winkelhaus, Grosse, and Morana 2021). They can support or replace tasks while enhancing staff's physical and cognitive processes (Grosse 2024), which is particularly relevant given the repetitive nature of tasks of OP (Dregger et al. 2018). These technologies also enable a more detailed characterisation of tasks, addressing oversimplifications caused by limited information. Previously, workers were considered homogeneous, with no attention to breaks, fatigue, or task learning (De Lombaert et al. 2023). Accordingly, digital technologies are indispensable for

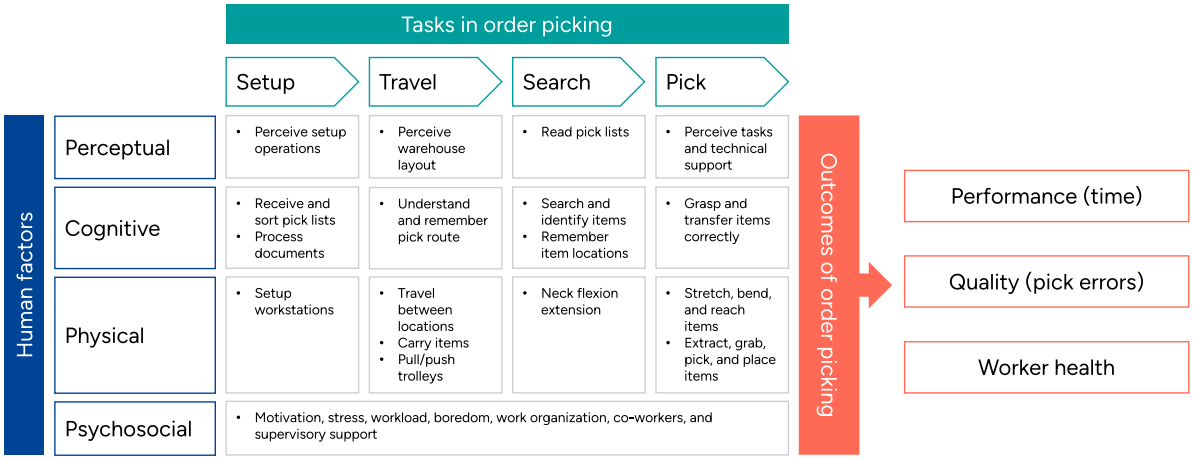


Figure 1. Human factors, tasks, and outcomes of order picking according to Grosse et al. (2015).

generating and analyzing information about picking, setup, travel, and search tasks in OP (de Koster, Le-Duc, and Roodbergen 2007; Glock et al. 2017; Vijayakumar et al. 2022). Setup involves administrative tasks before a pick tour, travel covers movement between storage locations, search refers to identifying parts on shelves, and picking includes collecting, documenting, verifying, and restocking items. Table 1 provides an overview of studies in OP applying digital technologies. The choice of the studies in the table is deliberate. It gives precedence to the recency and relevance of publications combining digital technologies and human factors in OP.

2.2. Reference architecture for enhancing human-centricity

Digital technologies are indispensable for achieving a human-centered approach in OP (Larco et al. 2017). These technologies enhance environmental awareness and facilitate the collaboration of humans and machines to improve the working conditions of staff through data-driven and interconnected systems (Dregger et al. 2018; Ghobakhloo et al. 2022; Grosse 2024; Lucchese and Mummolo 2024). Digital technologies are essential for generating, analyzing, and using real-time information that improve the tasks of staff and its outcomes (Enang, Bashiri, and Jarvis 2023; Gladysz et al. 2023; Sgarbossa et al. 2020; Winkelhaus and Grosse 2020).

This study applied the human-centered manufacturing architecture proposed by Rožanec et al. (2023). This was selected for its integration of privacy measures and task adaptability, which lack in existing architectures (Cimini, Lagorio, Pirola, et al. 2021; Guo et al. 2024; Konstantinidis et al. 2023; Leng et al. 2023; Moghaddam et al. 2018; Pivoto et al. 2021; van Geest, Tekinerdogan, and Catal 2021). It enables the development of trustworthy AI models that incorporate human input while ensuring data confidentiality, integrity, and availability in manufacturing. These capabilities are crucial for addressing information vulnerabilities and establishing security policies, key challenges in applying AI-enabled vision systems for staff monitoring (Generosi et al. 2022; Gupta and Prabhat 2023). Additionally, the architecture enhances task adaptability by generating information on staff actions, which is indispensable for non-intrusive data collection in AI-enabled vision systems (Slama et al. 2024). The architecture comprises seven layers, starting with Layer A, which represents the physical components of a system. Layers B through F focus on data acquisition, management, processing, and analytics, employing digital technologies and AI to transform real-world data into actionable insights. Finally, Layer G focuses on data visualisation and user interfaces to facilitate interaction between users and digital environments. Figure 2 presents the reference architecture for human-centered manufacturing systems according to Rožanec et al. (2023) and details the content of each layer.

2.3. Artificial intelligence-enabled vision systems for information generation and task analysis

Advances in AI-enabled vision systems contribute to generating and analyzing information of tasks (Konstantinidis et al. 2023; Zhou et al. 2023). These systems, when applied to the tasks of staff, can identify actions, resources, or intent based on historical data and present suggestions for improvement (Benmessabih et al. 2024). Research suggests that their contribution in a human-centered approach may be critical as they potentially integrate human factors including physical (Bortolini et al. 2020), psychosocial (Fang, Zheng, and Xu 2017), perceptual (Helm, Malikova, and Kembro 2023), or cognitive (Lou et al. 2022). One AI method favoured by

Table 1. Digital technologies and human factors of order picking.

Study	Topic	Review	Digital technologies	Human factors	Tasks
Grosse (2024)	Supportive and substitutive technologies in manual warehouse OP	**Yes**	Information generation: Tablets, wareable devices, pick-by-voice, wearable devices, automated guided vehicles (AGV), collaborative robots, and autonomous mobile robots	Cognitive, physical, and psychosocial	Setup, travel, search, and pick
Rojas et al. (2024)	Digital technologies for supporting staff with disabilities identifying gaps and barriers in literature	**Yes**	Information generation: collaborative robots, wearable devices, exoskeletons, projection, augmented reality, and motion recognition	Physical, perceptual, cognitive, and psychosocial	Setup, travel, search, and pick
Glock et al. (2021)	Technical assistive devices for material handling operations in warehouses that reduce human workload	**Yes**	Information generation: RFID, wareable devices, pick-by-voice, vision, and tablets	Cognitive, physical, and psychosocial	Setup, travel, search, and pick
Neumann et al. (2021)	Negative consequences of ignoring human factors and proposed a framework for including human factors when applying digital technologies	**Yes**	Information generation: RFIDs	Perceptual	Setup and pick
Winkelhaus, Grosse, and Morana (2021)	Digital technologies in OP including the intersection of social and technical aspects	**Yes**	Information generation: RFIDs, IIoT devices, wearable devices, and AGVs	Cognitive, physical, and psychosocial	Setup, travel, search, and pick
Sgarbossa et al. (2020)	Challenges and opportunities for integrating human factors in production and logistics including the use of digital technologies	**Yes**	Information generation: Motion capture systems, cyber physical systems, and digital twins	Perceptual, cognitive, physical, and psychosocial	Setup, travel, search, and pick
Lucchese and Mummolo (2024)	Human-centric outcomes of cognitive assistive technologies supporting staff in tasks of OP and assembly	**No**	Information generation: Pick-by-light and scan, and tablets	Perceptual, cognitive, and physical	Setup, travel, search, and pick
Lucchese, Panagou, and Sgarbossa (2024b)	Performance of staff with different ages using digital technologies for OP	**No**	Information generation: pick by scan, pick by light, tablet, camera, and motion recognition	Perceptual and cognitive	Setup, travel, search, and pick
Lucchese, Panagou, and Sgarbossa (2024a)	Digital technologies that support staff in OP and assembly while assessing performance and well-being	**No**	Information generation: pick by light, pick by scan, pick by voice, and augmented reality	Perceptual and cognitive	Setup, travel, search, and pick
Loske et al. (2024)	Design of OP and cognitive demands impacting the performance of staff	**No**	Information generation: tablet	Perceptual and cognitive	Setup, travel, search, and pick
Hosseini et al. (2024)	Impact of automated guided vehicles on boredom and performance of staff in warehouses	**No**	Information generation: tablet, smart phone, automated guided vehicle	Psychosocial	Setup, travel, search, and pick
De Lombaert, Braekers, et al. (2024)	Choice of orders by staff in warehouses and effects on productivity	**No**	Information generation: Wearable devices, warehouse management system	Perceptual, cognitive, and psychosocial	Setup, travel, search, and pick
De Lombaert et al. (2023)	Requirements, human-centric planning design, and human factors in OP	**No**	Information generation: Wearable devices, monitoring devices, exoskeletons, cobots, pick-by-voice, and vision systems	Physical	Setup, travel, search, and pick
Pasparakis, Vries, and Koster (2023)	Consequences of human-robot collaboration on behavioural outcomes and staff turnover in OP	**No**	Information generation: Autonomous mobile robots	Cognitive and psychosocial	Setup, travel, search, and pick
Falkenberg and Spinler (2023)	Machine learning to integrate tasks and human factors of OP and predict productivity	**No**	Information analysis: ML including extreme gradient boosting for planning tasks of OP	Physical and psychosocial	Travel and pick

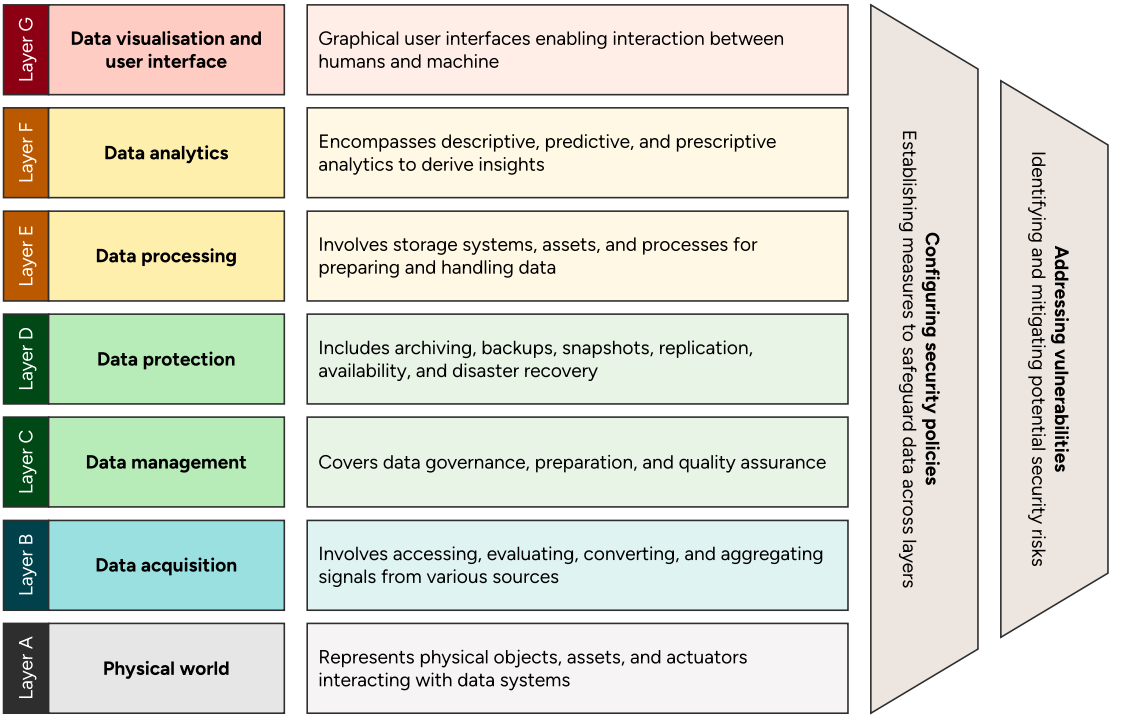


Figure 2. Architecture of human-centered manufacturing based on Rožanec et al. (2023).

the literature is machine learning (ML), which can collect data and identify patterns in images and videos including the movement of materials and staff (Lou et al. 2022). This includes the use of artificial neural networks (ANNs), which involve interconnected neuron-like units to process information (Shahin et al. 2024; Verdhan 2021). Trained for specific tasks, these networks progressively analyze visual data, and require training sets to learn and verify patterns (Ansari 2023). The output is compared with a human-provided labels (e.g. gestures, movement, pose, or interactions of staff with its environment). Appendix 1 describes the rationale for the selection of ANNs and presents the specific algorithm applied in this study.

Currently, the literature lacks studies that apply AI-enabled vision systems to generate and analyze staff-related information in OP (Helm, Malikova, and Kembro 2023; Konstantinidis et al. 2023). It is therefore essential to extend existing frameworks to include such systems. This study proposes two aspects for extending such frameworks. First, it conceptualises AI-enabled vision systems as integral digital technologies within a broader human-centered manufacturing architecture. Second, it asserts that applying these systems on OP must satisfy the logical steps necessary to realise AI-enabled vision applications in manufacturing environments (Konstantinidis et al. 2023; Li, Zhao, and Yang 2023; Zhou et al. 2023). These steps have been formulated by Smith, Smith, and Hansen (2021), and include five activities: image acquisition, pre-processing, segmentation, feature extraction, and classification or interpretation. Image acquisition involves capturing visual data using cameras under optimised lighting conditions (Torres et al. 2022). Pre-processing enhances image quality by reducing noise and improving clarity (Benmessabih et al. 2024). Segmentation partitions images into regions to detect and isolate relevant objects (Parsa et al. 2019). Feature extraction simplifies image data to facilitate pattern recognition (Penumuru, Muthuswamy, and Karumbu 2020). Finally, classification assigns labels using supervised, unsupervised, or semi-supervised learning methods (Konstantinidis et al. 2023; Lou et al. 2022). Figure 3 depicts these five activities of AI-enabled vision systems, as outlined by Smith, Smith, and Hansen (2021) and their potential application for OP.

2.4. Research gap in the state of art

The literature increasingly acknowledges the importance of a human-centered approach to OP. There has been a notable shift from theoretical model development toward empirical studies analyzing the human factors involved in OP tasks, with a focus on improving staff outcomes. The studies in Table 1 highlight the role of digital technologies in generating and analyzing task-related information that might otherwise go unnoticed, but that is essential to support staff well-being. Despite these advancements, the application of AI-enabled vision systems for

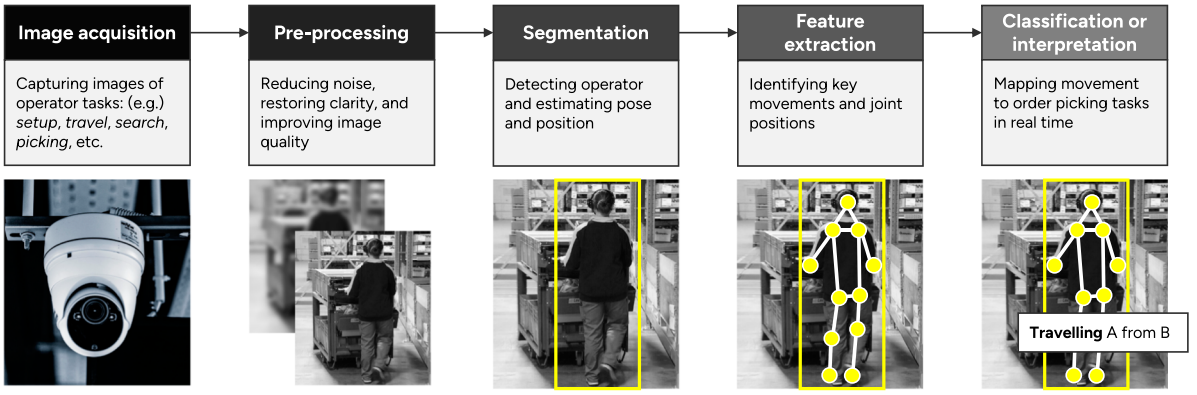


Figure 3. Five activities of AI-enabled vision systems according to Smith, Smith, and Hansen (2021).

data collection in human-centered OP remains underexplored. To bridge this gap, this study highlights the need of adopting human-centered architectures that integrate privacy measures, support task adaptability, and incorporate AI-enabled vision systems. In agreement with existing literature, this study maintains that applying AI-enabled vision systems in OP must be guided by a clear understanding of key image processing stages–namely, acquisition, processing, segmentation, feature extraction and classification of images. Accordingly, this study contributes to literature by answering the following research questions:

RQ 1 – ***How can AI-enabled vision systems contribute to generating and analyzing information in human-centered OP?***

RQ 2 – ***How can privacy concerns associated with AI-enabled vision systems used in human-centered OP be mitigated?***

3. Research design

This section outlines the research design of the study, including the choice of DSR, and a case in automotive manufacturing sector. Subsequently, it describes the steps of data collection and analysis.

3.1. Design science approach

The study involved industrial and academic partners collaborating to solve a practical problem and extending knowledge of AI-enabled vision systems in OP. It applied a DSR approach, which focuses on developing generic knowledge through discovery and technological problem solving (Holmström, Ketokivi, and Hameri 2009). The literature recommends applying DSR to practical problems that focus on the design of systems and implementation of actions to achieve a desired outcome in practice (van Aken, Chandrasekaran, and Halman 2016). Unlike other approaches of research, DSR creates artificial phenomena or artefacts to make a creative jump that results in reflections originating from changes to an existing system (March and Smith 1995). In contrast, Delphi studies rely on expert-based empirical data to build consensus and support exploratory theory development (Agrawal et al. 2024).

To structure the design and evaluation process, this study adopts the CIMO logic framework, which facilitates the analysis of interventions within a specific context and examines how underlying mechanisms produce intended outcomes. CIMO logic originates from realistic evaluation within design science, recognising the effectiveness of an intervention is shaped by contextual conditions, implementation processes, and the mechanisms driving change (Halminen et al. 2021). In this study, the context (C) refers to the environmental and organisational conditions that define the problem space. The intervention (I) involves the development of an AI-enabled vision system designed at generating and analyzing information related to OP tasks. The mechanism (M) encompasses the processes of testing and iterative feedback through workshops and pilot studies, ensuring active stakeholder involvement in refining the solution. The outcome (O) comprises enhanced task visibility, improved operational efficiency, and privacy-compliant monitoring, contributing to more human-centered OP practices. The CIMO framework guides the practical implementation of AI-enabled vision systems but also provides theoretical insights that extend beyond the demonstration of a technical solution. Figure 4 depicts how this logic structures the study's research questions and methodology.

3.2. Case description

The company in this case study is part of an international project on the use of AI-enabled vision systems to

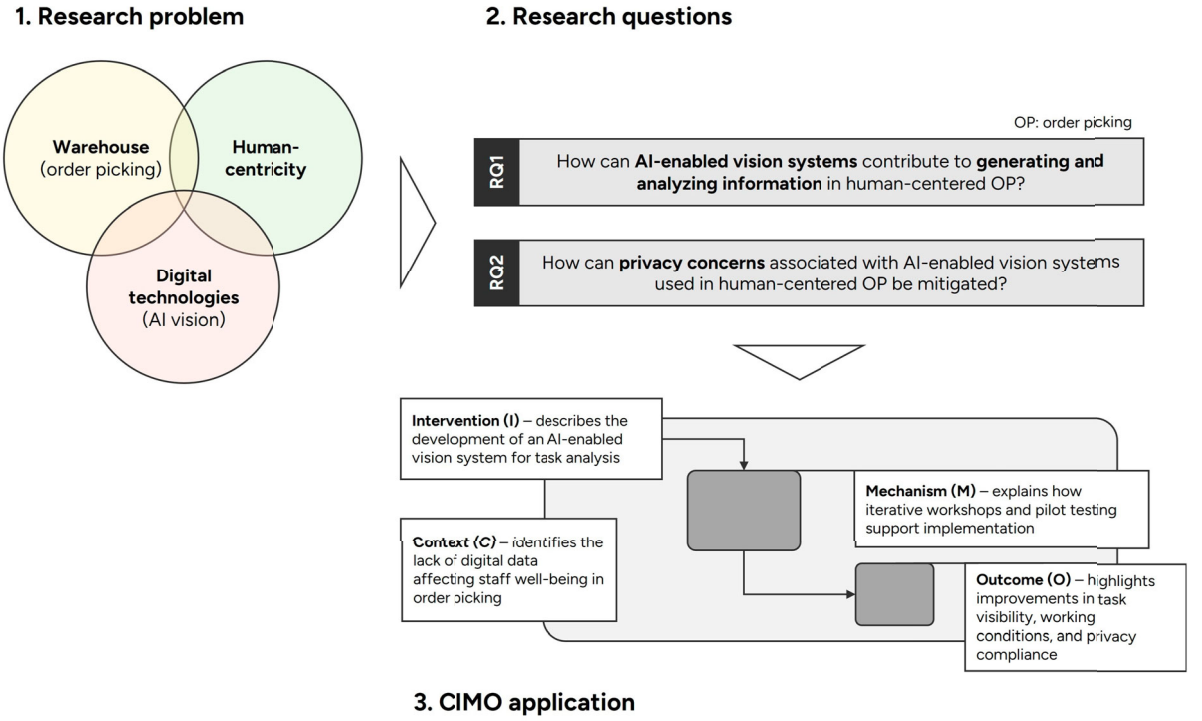


Figure 4. Research design process outlining the problem, research questions, and application of CIMO logic in a design science research approach.

improve logistics in automotive manufacturing, involving both industrial and academic partners. It focussed on a Swedish manufacturer of heavy trucks, buses, and engines. The company is considered a global leader in its field and employs approximately 50,000 people across more than 100 countries. OP tasks in this context remain predominantly manual, with technologies such as pick-by-voice, pick-by-light, or tablets available and in use, but their implementation is not yet widespread across all warehouses. Digitalization efforts exhibit varying levels of maturity and include real-time location systems, automated guided vehicles, QR codes, and camera-based part detection. The company seeks to enhance working conditions for staff responsible for these tasks. The responsibilities of staff in OP are diverse, and include searching for parts, using lifting tools and carts for transport, disposing of packaging, replenishing material, controlling for quality of picks and preparing kits for the delivery of parts to assembly lines. Improving staff well-being is a priority, as it directly affects timely order fulfilment, picking errors, and staff satisfaction. The absence of digital information to identify tasks that hinder well-being of staff remains a major challenge. Current methods at the company–such as manually recording and analyzing staff activities–are time consuming, rely on extensive engineering expertise, and are impractical for the scale of operations. This case presented an opportunity for applying AI-enabled vision systems to analyze the generation and analysis of information of tasks in OP for warehouses. Additionally, it provided rich data through continuous interaction with stakeholders, enabling a collaborative problem-solving approach.

3.3. Data collection

Data collection occurred between April 2023 and July 2024. The data collection was conducted to characterise activities for generating and analyzing information about OP when using AI-enabled vision systems. Additionally, it investigated privacy concerns of these systems. Data collection included four objectives. First, understanding the practical challenges of OP. Second, defining staff responsibilities and the expected outcomes of using AI-enabled vision systems. Third, developing a collaborative solution for generating and analyzing unstructured data while ensuring privacy. Fourth, applying digital technologies aligned with stakeholder needs in OP. Data collection included semi-structured interviews, field notes, and company documents. Additionally, it included the development of a pilot study in a laboratory environment between January and July 2024. Data collected during the pilot study prioritised the monitoring of staff tasks using an AI-enabled vision system during OP. These data consisted of geometric information, time, and labelled activities of staff (described in Section 4.2). Semi-structured interviews were chosen as prior studies that have used

this method to examine the interaction between OP tasks and human factors (Grosse et al. 2016). The research team consisted of one professor, two assistant professors, a postdoc researcher, a research engineer, two master's students, and stakeholders from the company. Company stakeholders included a logistics manager, an engineer overseeing OP, two engineers handling time-data management, a logistics developer, and a digitalisation manager. Additional stakeholders included a project manager and the technology director of a supplier participating in the project and specialising in vision-based systems for logistics. Sixteen staff members were interviewed during their shifts while they performed tasks of OP. Interviewees were purposively selected to capture diversity across age groups, gender, professional experience, cultural backgrounds, and language. The group included 13 male and three female employees with diverse backgrounds from Europe and Asia. Their average age was 39,75 years with a standard deviation of 9,09 years. The height of employees ranged between 1,65 and 1,90 m. Their average experience was 3,21 years with a standard deviation of 3,48 years. Each was employee typically responsible for up to five warehouse aisles. Field notes originated from six full day study visits at three warehouses of the company involving OP. Finally, the collected information included company documents specifying the tasks, order, and time data analysis of OP.

3.4. Data analysis

This section outlines the steps for processing the collected data following the CIMO logic in a DSR approach. The context (C) identified the problem of lacking digital data for detecting tasks hindering staff well-being in OP. This included understanding current methods for collecting data about OP, staff activities, digital solutions, and human factors considered by the company. Subsequently, interviews were conducted with engineering, managerial, and digitalisation staff in logistics, followed by interviews with staff responsible for OP. Appendix 2 presents the interview protocol. This provided insights into the tasks of OP (setup, travel, search, and pick), human factors (perceptual, physical, mental, and psychosocial), and outcomes (performance, quality, and worker health), as depicted in Figure 1. Finally, the data were analysed and the findings were presented to company stakeholders, including the research and development manager and logistics staff, in a project meeting.

For the intervention (I), the study characterised the activities for generating and analyzing information about tasks of OP that enhanced human-centricity. The research team, in collaboration with six company stakeholders, iteratively developed the intervention over the course of three workshops, focussing on key aspects of AI-enabled vision systems. Consistent with previous studies, the intervention prioritised stakeholder engagement to improve the validity and practical relevance of academic findings (De Lombaert, Braekers, et al. 2024). The workshops focussed on identifying relevant human factors, tasks, and outcomes (Grosse et al. 2015) and analyzing the use of a human-centered manufacturing architecture (Rožanec et al. 2023). Based on this, the research team developed an architecture for an AI-enabled vision systems including physical components, data flow, information processing, algorithms, and a development plan for testing in a laboratory environment.

The mechanism (M) of this study involved developing and testing AI-enabled vision systems to improve human-centric OP. Based on the collected data, a pilot study was conducted in a laboratory environment to generate and analyze information related to OP. This was conducted by two assistant professors, a postdoc researcher, and two master's students. The company provided industrial hardware, whereas the supplier provided cameras, software, and technical support. Iterative testing included setup, calibration, model development, and training of the AI-enabled vision system, conducted in collaboration with the supplier and its technical staff. Feedback from company stakeholders was gathered during bi-weekly meetings, in which preliminary results were presented to the digitalisation manager and logistics developers.

The outcomes (O) of the study comprehended the activities for applying ML to identify patterns in unstructured data originating from OP. To achieve this, the study followed the steps of image acquisition, preprocessing, segmentation, feature extraction, and classification suggested by Smith, Smith, and Hansen (2021). The research team reviewed the results of generating and analyzing information about tasks including setup, travel, search, and picking when using AI-enabled vision systems. The classification process involved visualisation, feature identification, manual labelling, and subsequent training and testing of a ML model to automatically classify OP tasks. The study evaluated the following features for task classification: staff ID, timestamp, ankle midpoint, geographical location, staff speed, and wrist position. The algorithm prioritised consistency in task identification when compared to manual labelling, serving as a benchmark for performance evaluation. Additionally, the team described the implications for managing privacy concerns when monitoring the tasks of staff. These findings were presented in three meetings to company stakeholders and its supplier. Thereafter, the research team discussed the implication of these results to human factors including perceptual, physical, mental, and psychosocial

factors. The discussion highlighted the contribution of these factors to outcomes like performance, quality, and worker health.

The study applied three measures to improve the validity and reliability of its results. First, it applied distinct types of data collection to triangulate information and enhance the validity of the study. Second, the study prioritised field testing to achieve pragmatic validity, with a focus on the future application of AI-enabled vision systems for OP. Third, the choice of a case was based on theoretical sampling, providing a rationale for case selection. This was performed to compare the findings of the study with previous ones and enhance the generalizability of its findings. To enhance the generalizability of its findings, the study focussed on key characteristics of warehouses that are critical to human-centered OP. These included the transportation of parts, kitting, strict time constraints, repetitive travel patterns, and consistent part presentation. Such features are essential for representing warehouses as socio-technical systems, regardless of industry type (Winkelhaus, Grosse, and Morana 2021). These systems comprise staff (the social component), digital technologies (e.g. AI-enabled vision systems), and the physical workspace, all of which interact and influence one another (Dregger et al. 2018). This focus is crucial because adopting a socio-technical perspective is necessary when implementing digital technologies in OP–one that not only prioritises productivity but also considers the well-being of workers.

4. Applying CIMO logic and empirical results

This section presents the empirical findings from the application of an AI-enabled vision system in OP. The empirical results follow the CIMO logic in DSR.

4.1. Context

The company considered the generation and analysis of information of tasks a critical aspect for considering the human factors and performance of OP. Logistics engineers identified that the lack of digital information hindered the detection of poorly designed tasks, contributed to uneven work pace resulting in stressful scenarios, and affected the design of the warehouse with hard-to-reach and hard-to-find locations. Information about tasks was based the assumption of standard times that considered staff homogeneous. The identification of problems was ad hoc, and originated from the notification of repeated complaints by staff, such as delayed deliveries. To address these problems and enhance a human-centered approach to OP, the company required increased knowledge about the tasks of staff related to OP, its time period, and the detection of tasks affecting human factors. The company considered the integration of AI-enabled vision systems with existing technologies (e.g. pick-by-voice, pick-by-light, and tablets) to improve understanding of OP tasks and their outcomes. Table A1 in the Appendix presents the negative working conditions identified as affecting staff well-being in OP.

The picking process involves the delivery of parts to the assembly line every 16 min, with up to 20 articles for every picking order. Examples of these parts include gearboxes, disc breaks, and screw packs for different products and product variants. Staff use lifting tools, manual carts, radio communication, and visual aids to maintain the pace of OP. Paper based solutions are common, and only one warehouse has implemented the use of tablets. Tablets present information for OP in real-time and communicate with enterprise resource planning (ERP) systems indicating changes and ensuring the right amount and type of products are picked. Maintaining the pace of picking requires significant effort from staff. Tasks in OP are uneven throughout the day, causing stressful moments. Experienced staff locate parts easily but those with less experience must walk extensively to locate parts. Staff invest significant time in the process of validating the quantity and type of items for every product in every picking order.

Picking begins by presenting staff with orders in both printed form and on a screen. Information about the order originates from an ERP system. These orders specify the parts to be picked, their quantities, and their locations in the warehouse. Staff travel to various warehouse locations with a cart to collect the parts. Upon locating a part, they use lifting tools to retrieve and place it in carts, verifying the quantity and quality of parts. This process repeats until all parts on the list are picked. Staff then confirm the order's completion in the ERP system and place the cart in a designated area for collection. In the remaining time, they prepare materials for the next sequence, communicate with material handling staff to resupply depleted parts, and handle tasks such as removing packaging and adjusting picking sequences.

Regarding the generation and analysis of tasks, engineers have limited visibility of the tasks of staff in OP, only identifying issues when staff report them. Engineers then manually record tasks with hand-held cameras, analyze them, and discuss potential improvements with OP staff. The ratio of data recording to analysis for identifying problems about human factors is one minute of recording to 30 min of analysis. In its analysis, tasks of OP are categorised by company-defined time blocks, which include standardised tasks applied across the factory to establish OP standards and detect issues affecting staff or performance. The company anticipates

Table 2. Resources for order picking in intervention.

Resource for order picking	Description
Task definition for staff	**Setup**: material replenishment, order preparation, disposing of packaging, retrieving orders, gathering physical equipment, order confirmation. **Pick**: lifting, retrieving, grabbing, and placing products in material cart. **Travel**: any staff movement including with or without material. **Search**: reading OP lists, identifying items or material location
Information for execution of OP	Part number, location, code, description, picking method, OP number
Resources and materials for OP	Lifting tool, tablets, OP list, material cart, products, packaging, material disposal, mobile robot, warehouse areas, picking bins

an increase in production rates and product diversity, which could worsen staff conditions under the current practices. Therefore, developing human-centered solutions that balance staff well-being and operational performance is critical, and generating and analyzing unstructured data from picking is a key first step. Table A2 in the Appendix outlines the problems identified in the company when generating and analyzing information and privacy concerns.

4.2. Intervention

To address the problem, the study characterised the components, rules, constraints, interconnection and interactions of an AI-enabled vision system. The intervention began by defining the resources or physical components to create an AI-enabled vision system in a laboratory environment. These included defining the tasks of the staff, the information for execution of OP, and resources and materials for completing tasks. Table 2 describes the resources for OP in the intervention. Then, it described the functions or activities performed by the AI-enabled vision system. To achieve this, the study gave precedence to specifying the activities for generating and analyzing information in AI-enabled vision systems suggested by Smith, Smith, and Hansen (2021). Completing these activities required consideration of different algorithms for increasing the quality of images, portioning visual information, detecting staff, extracting salient features, and labelling data. Next, it specified the integration of physical and functional components of an AI-enabled vision system. This included digitising staff activities (e.g. setup, travel, search, pick) and their interactions with information such as OP lists, resources, and materials.

The intervention specified the components of a seven layer architecture (Layers A to G) suggested by Rožanec et al. (2023) including physical components, processing of data, data management, data protection, data processing, analytics and visualisation.

The implementation of the physical world (Layer A) replicated the company's OP process. It included 19 parts stored in a warehouse with varying stock levels, as well as designated areas for a material handling cart, package disposal, and completed orders. The system utilised a material handling cart, a lifting devices, and a mobile robot. Picking lists specified product specifications, and orders were presented either on paper or tablets. Staff collected items, placed them in the cart, and transferred completed orders to the mobile robot, which transported them to the assembly station. A ceiling-mounted camera monitored operations, with its placement optimised to reduce occlusion. Data acquisition (Layer B) was performed using a two-dimensional camera and a computing unit responsible for visual data processing. Challenges included camera calibration, requiring measurement and marking of warehouse features, and ensuring consistent data transmission through improved lighting, stable internet connectivity, and a robust data pipeline. Data management (Layer C) involved the creation of a database to store OP information, ML model inputs/outputs, and error reports. Key challenges included managing high data volumes, structuring the information, and ensuring efficient retrieval. Solutions included the use of cloud-based storage, structured file naming, and database management.

To comply with the European Data Protection Supervisor (EDPS) recommendations (Layer D), data collection was minimised to relevant areas and initiated only during OP. Staff were notified before each session, and video footage was not stored. Instead, positional coordinates without personal identifiers were used. Privacy measures included facial blurring, restricted data access, and secure local hosting with password protection. Challenges included increased data availability potentially exposing sensitive information, which could involve data leakage, unauthorised use, data modification, and improper access. Gaining the trust and acceptance of staff required identifying jointly concerns about data protection and privacy, and demonstrating how actions addressed these concerns.

Data processing (Layer E) relied on middleware to coordinate APIs, hardware components, and cloud services. The system captured positional data of staff to inform them about order requirements, support ML-based task classification, and trigger mobile robot operations for item transport. Data analytics (Layer F) involved image acquisition, pre-processing, segmentation, feature extraction, and classification using AI-enabled vision

systems. Among these, task classification emerged as the most complex challenge. The user interface (Layer G), deployed on a tablet, supported the OP process by displaying order lists, part details, confirmation options, staff authentication, and error reporting. Iterative refinements were made to enhance usability and effective visual design.

4.3. Mechanism

Testing conditions of the pilot study included tasks of OP and generation and analysis of information using AI-enabled vision systems in a laboratory environment. It exemplified the industrial case and limited the warehouse area to a height, length, and breadth of 2,10 and 3,5 m. Testing began with staff retrieving a picking list and collecting a material handling cart. They then travelled through the warehouse to locate and pick the listed parts. After each part was picked, staff verified the quantity and quality, one part at a time. Next, they transported the cart to the area for completed orders, confirmed the order's completion, and verified the parts again. Staff then called the mobile robot to collect and deliver the parts to the assembly area. After handing over the parts, they returned the cart to its starting position, concluding the process. This cycle repeated until all orders were completed, with the picking pace matching that of the assembly station.

Two OP characteristics highlighted by company stakeholders were tested in the pilot study. First, heavy items required the use of a simulated lifting tool, mimicked by a block and tackle system, which staff operated to simulate an industrial lifting tool. Second, the picking lists included both frequent parts, required for all products, and infrequent parts, required for special features. Additional activities included disposing of packaging and replenishing parts in the warehouse. Staff discarded packaging when opening new boxes and replenished parts after OP when a location ran out of stock. Throughout these activities, the AI-enabled vision system continuously acquired data. Figure 5 presents the mechanism of the study including the physical world of AI-enabled vision systems for OP.

The mechanism implemented the five activities for generating and analyzing information, as outlined in Section 2.3. These activities included image acquisition, pre-processing, segmentation, feature extraction, and classification. Image acquisition involved positioning the camera to ensure a complete view of the picking equipment, staff, and aisles during OP. The camera placement was carefully selected to minimise obstructions from equipment or staff and to ensure adequate lighting conditions throughout task execution.

Pre-processing improves the quality of images by enhancing, restoring, transforming, detecting edges or corners, registering or preparing images for training. This study limited pre-processing to normalising images to prepare these for the training of a ML model. Consistent light conditions, absence of occlusions, the presence of only one operator, and the need for accessing a full data set to later segment data explained this situation.

Segmentation focussed on detecting staff in the laboratory environment, and verifying whether the same individual remained present throughout the OP task. It also involved estimating the position of staff and predicting their future position based on its current one. These steps were essential for accurately associating operator activities with specific tasks, resources, and items on the picking list. Two ML models including a Hungarian algorithm and Kalman filter were applied. The model results provided the geometrical positions of 17 human key points in a time series.

Feature extraction involved isolating the most informative elements from the original data set. A Python script (Appendix 1) was used to evaluate and visualise the time-series data of the 17 human key points obtained during segmentation. This analysis showed that only five features were necessary to consistently identify staff in the pilot study, which included the left and right ankles, left and right wrists, and speed of staff.

The classification of tasks was conducted in two steps. First, the data visualisation and the saliency of selected features were used to manually label OP tasks. Manual labelling of data was based on ten rounds of OP selected randomly, including distinct staff. The label of pattern classification referred to the consistency in the detection and movement of staff in a zone. The label of activity classification identified the tasks of OP including pickup, search, travel, and setup. The label of specific activity distinguished the picking of parts from the upper and lower shelf of the warehouse. Figure 6 presents an example of the visualisation of features used for manual labelling of OP data in the application of AI-enabled vision systems. Second, it applied a ML algorithm (XGBoost) to train and test a model that classified tasks of OP. Figure 7 shows the information generation and task labelling process using the AI-enabled vision system.

4.4. Outcomes

The outcomes of the study focussed on identifying the key aspects of the generation and analysis of information and privacy when using AI-enabled vision systems in OP.

Stakeholders from the company and its supplier discussed how AI-enabled vision systems enhance information generation for OP. They identified three key benefits.

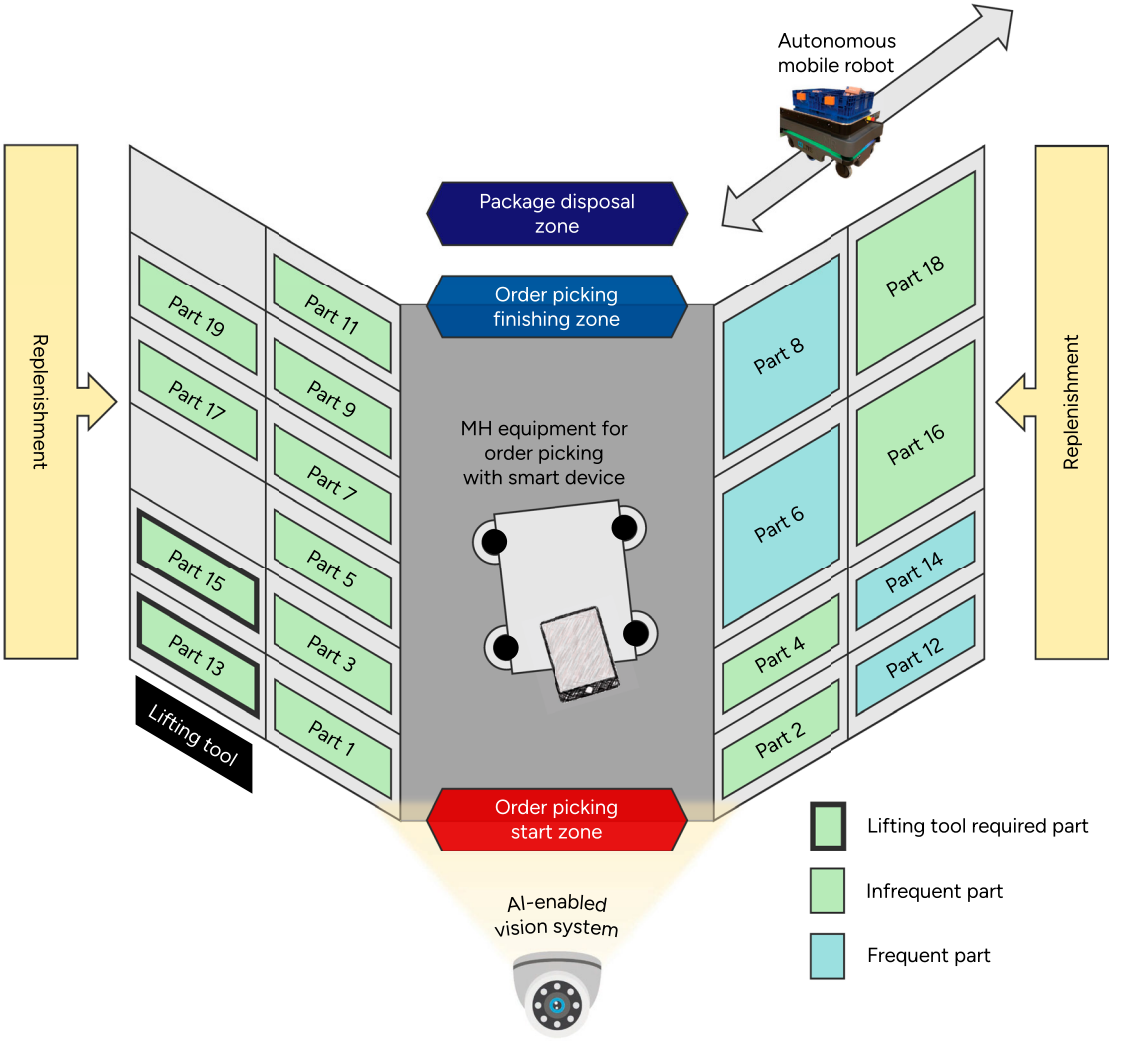


Figure 5. Mechanism including the physical world of AI-enabled vision systems for order picking.

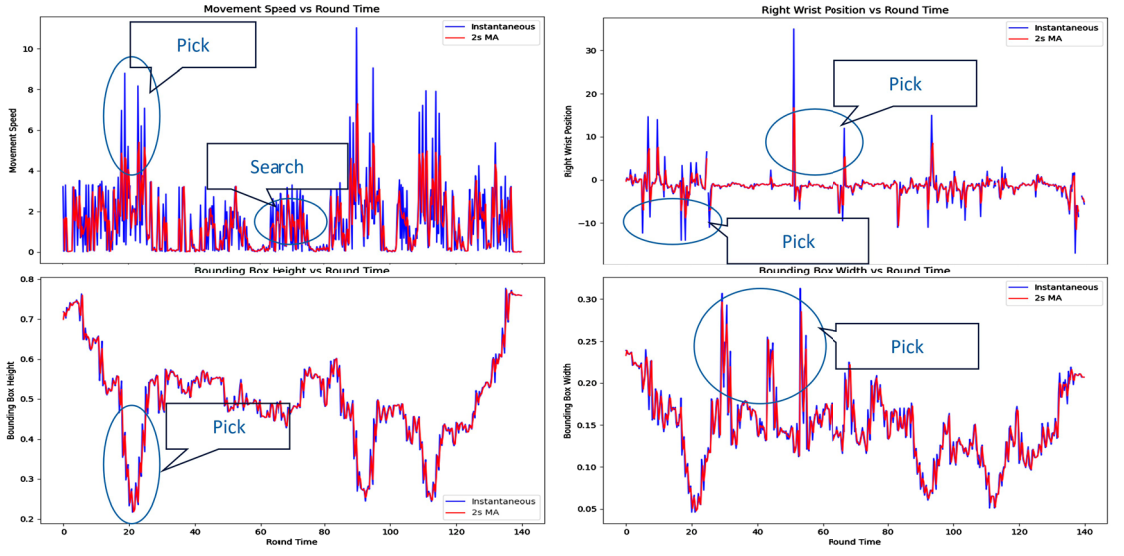


Figure 6. Example of visualisation of features for manually labelling data in order picking when applying AI-enabled vision systems.



Figure 7. Example of labelling of tasks when applying AI-enabled vision system to order picking.

First, the system continuously generated information on OP. This eliminated the need for lengthy observations, inconsistent data collection, task interruptions, and unnatural task execution (e.g. staff altering speed) typical of manual recording. This was noted as crucial for reducing resources and costs related to the analysis of OP. Second, the system generated information based on interactions among staff, tasks, parts, equipment, and aisles, enhancing transparency and visibility of tasks in OP. This information supported the development of frequency distributions and the identification of patterns and outliers in the execution of tasks of OP, which was considered essential for improving operations of warehouses based on data. Third, the system generated and stored information digitally, which could be accessed, shared, processed, or used by staff or digital services. This was considered important for reducing the time spent searching for information on OP, eliminating paper-based practices, improving analysis of operations, and having a common dataset that facilitated sharing of knowledge.

Three advantages of AI-enabled vision systems for analyzing information in OP were described by company stakeholders. First, the system classified tasks into setup, search, travel, and pick based on unstructured data originating from visual monitoring in the warehouse. Engineers considered this advantageous when compared to current practices relying on manual collection and analysis of OP information. Current practices are time consuming and require scheduling resources, time, and staff during daily production for monitoring OP tasks.

Engineers noted that manual monitoring can be stressful for OP staff, who perceive it as a form of performance evaluation. They emphasised that AI-enabled vision systems offer an advantage by capturing the natural working conditions of staff, who might alter their behaviour during manual monitoring. Second, the system improved problem detection in OP. This information supported the development of frequency distributions and the identification of patterns and outliers in the execution of OP tasks, which was considered essential for improving operations of warehouses based on data. It highlighted tasks taking longer than expected, suggesting areas for improvement. For example, extended time spent searching for materials, discarding packaging, or replenishing products indicated inefficiencies in warehouse design or the need for modifying processes. Additionally, engineers viewed the use of AI-enabled vision systems as beneficial for standardising OP monitoring and reducing reliance on individual judgment. This was important, as engineers responsible for monitoring OP tasks were often overwhelmed by video analysis, had limited time, and struggled to consolidate information and reach consensus when interpreting OP information. Third, the system increased visibility and transparency across all tasks of OP. Stakeholders noted that they previously prioritised the tasks of picking and travelling, whereas search and setup tasks were often assumed to be minor or handled by staff other than that of OP. The system revealed that a comprehensive view of all tasks was essential and, unlike previously thought, searching and setup were particularly important for staff unfamiliar with OP or the laboratory environment. Engineers argued that improved visibility and transparency in OP could support better decision-making, such as identifying problematic areas in warehouses or inefficient OP tasks. It could also enhance the ability to detect, communicate, and respond rapidly to issues.

The use of AI-enabled systems for generating and analyzing information in OP was not without challenges. The first challenge involved generating information with the system after changes to the warehouse environment, such as the introduction of new parts, routines, staff, layouts, equipment, or digital technologies (Kadir and Broberg 2020). Stakeholders noted that the system might fail to recognise, classify, or acknowledge new information when this differed from the existing trained dataset. Additionally, the system provided erroneous data when changing the position of the camera, a scenario that required re-calibrating the system. The second challenge resulted from changes affecting image acquisition from the vision system. These included objects obstructing the camera, lightning changes, image blurring or shifts of reference points that determined the camera's coordinate system. These resulted in erroneous or unusable data. The third challenge concerned the time-intensive manual labelling of tasks by domain experts. Labeling was complicated by task variation, as the same task might be performed differently by the same or different individuals. Staff might also execute multiple tasks in parallel, such as travelling, picking, pausing, or returning an item while searching for parts during OP. This was critical as manual labelling affected the training of the model and eventually the classification of tasks. Despite these challenges, stakeholders argued that the benefits of AI-enabled vision systems may outweigh the drawbacks. Although no numerical cost benefit analysis was conducted, stakeholders agreed that monitoring and improving OP tasks is highly resource intensive. Most current resources are devoted to problem detection, with comparatively little time spent on developing OP solutions. Stakeholders noted that AI-enabled vision systems could facilitate the reallocation of resources toward creative problem-solving and the development of multiple solution alternatives. This shift could help transform existing OP analysis practices, which often prioritise cost effectiveness over operator well-being.

Addressing privacy concerns in AI-enabled vision systems for OP was a key focus for the company stakeholders. They emphasised that building trust in these systems was essential and outlined several actions to achieve it. First, they highlighted the importance of explaining the system's purpose and obtaining staff consent, which included clarifying the existing challenges that led to the system's implementation, expected outcomes, equipment type and capabilities, and the system's functionality. The stakeholders emphasised the importance of explaining how data generation and analysis would be conducted, including measures for data anonymization and confidentiality. Involving staff at all seniority levels–such as operators, engineers, and managers–was considered essential, which would enable them to decide on key aspects like camera placement, field of vision, and tasks included in analyses. Printed signs indicating camera presence in the warehouse was also deemed fundamental. Regarding the architecture of the AI-enabled vision system, company stakeholders described the importance of establishing mechanisms for generating anonymized data. For example, capturing task execution variations without linking data to individuals. The stakeholders highlighted that anonymized data avoided assumptions that ignore individual differences such as age, experience, and workload. Providing collective feedback based on anonymous data, rather than attributing actions to individuals, was also considered vital to fostering

staff engagement, supporting system acceptance, maintaining privacy, and avoiding counterproductive micromanagement.

5. Discussion

This section discusses the empirical results of the study and answers its research questions. It describes the use of AI-enabled vision systems to generate and analyze information in human-centered OP (RQ1). Then, it examines actions for mitigating privacy concerns associated with the use of these systems (RQ2). Next, it analyzes the implications of AI-enabled vision systems to human factors in OP and presents recommendations to managers. Finally, it outlines the limitations of the study and provides future directions of research.

5.1. Generating and analyzing task information (RQ1)

In response to RQ1, the study found key steps for monitoring tasks in OP when using AI-enabled vision systems, identified salient challenges and proposed activities for enabling its use. This study found that AI-enabled vision systems in OP rely on pattern recognition and the visual interpretation of unstructured data. These systems can classify tasks, measure task durations, and track task sequences, resource usage, and staff locations within the warehouse. For example, time spent on picking, setup, travel, and searching was linked to specific warehouse zones, picked parts, and tools, enabling the identification of the most time consuming activities. The study showed that AI-enabled vision systems can classify OP tasks based on time and offer a valuable method for generating data to improve process visibility. This is significant because time constraints in OP is known to increase stress, reduce quality, and affect performance and well-being, yet the specific tasks contributing to these effects remain underexplored (De Lombaert et al. 2023). The results of the pilot study (Appendix 4) were encouraging and aligned with previous studies that relied on manual observation of OP tasks (Tompkins 1996). Despite this, the pilot study did not directly link time to OP outcomes. However, it did reveal time consuming tasks that pose risks to staff well-being, which constitutes a long standing issue noted in research (Winkelhaus, Grosse, and Morana 2021). Additionally, the study demonstrated that AI-enabled vision systems can generate comprehensive datasets that cover all OP tasks over time and across the full workforce. This result is important as it supports the notion of statistical datasets that account for factors influencing the well-being of a diverse workforce, rather than prioritising efficiency at staff expense (De Lombaert, Rijal, et al. 2024). The findings highlight that classification of tasks remains challenging, even when staff reach consensus on the type of tasks and use of AI-enabled vision systems. Staff in OP use varied approaches to task execution. This variation in task execution increases the resources required for accurate dataset labelling and training of classification algorithms.

The results indicated that AI-enabled vision systems can address two key challenges in a human-centered approach to OP. The first challenge involves deviations in work behaviour, where staff actions diverge from prescribed procedures due to autonomy, personal objectives, colleague influence, and the warehouse environment (Glock et al. 2017). This behaviour is frequent, spontaneous, and difficult to detect, and has an impact on staff well-being, lead times, service levels, motivation, and accident rates (De Lombaert, Braekers, et al. 2024). The findings suggest that AI-enabled vision systems can continuously monitor and classify tasks and potentially detect deviating work behaviour over time. This capability can allow managers and staff to collaboratively discuss corrective actions, prioritise solutions, and utilise AI as a tool to support task improvements. The second challenge is the lack of personalised training sessions in OP. This issue frustrates staff who lack the necessary knowledge to perform tasks and increases hiring and retention costs of warehouses (Kadir and Broberg 2020; Lucchese, Panagou, and Sgarbossa 2024a). AI-enabled vision systems can continuously collect data on working conditions, including how staff approach picking locations or apply manual techniques. This information could support training customised to diverse staff profiles based on age, gender, and language, helping workers identify risks specific to their needs rather than relying on generic or biased approaches (De Lombaert et al. 2023). Integrating AI-generated data into training programs could improve staff retention and attract new workers, a concern frequently highlighted in research on human-centered OP (Loske 2022; Philips, Huang, and Bodine 2024). The results further suggest that AI-enabled vision systems can contribute to satisfying these challenges by generating datasets that capture the full range of task execution across individuals while preserving anonymity. This capability is essential for restructuring task management in OP based on real-time data, rather than on standard times that assume uniform staff characteristics.

The results of the study highlight the need to include the five activities of AI-enabled vision systems in the seven layers (A to G) of human-centered architecture as follows. Layer A defines the physical elements of OP including staff, materials, resources, and layout. In an AI-enabled vision system, this requires specifying the

subject of study and its interactions. Layer B consists of data acquisition. This corresponds to the activity of image acquisition in AI-enabled vision systems. It concerns structuring the analysed environment, including positioning of objects and selecting a camera, lens, and lightning conditions. Layer C includes data management, which corresponds to the activity of pre-processing in AI-enabled vision systems. This requires the use of algorithms to improve the quality of images that improve, reconstruct, or remove noise and may involve enhancement, restoration, or transformation techniques. Layer D refers to data protection. This corresponds to activities associated with enhancing privacy in AI-enabled vision systems. Such as informing staff about the purpose of data acquisition, limiting the scope when generating information, achieving obfuscation, anonymizing metadata, restricting access, or hosting applications and storage in secure servers. Layer E involves data processing, which corresponds to the activities of segmentation and feature extraction in AI-enabled vision systems. Segmentation involves partitioning visual information and the detection of objects. Feature selection involves acquiring significant information from high dimensional information. Layer F consists of data analysis, which corresponds to the activity of classification or interpretation in an AI-enabled vision system. This activity involves assigning labels to extracted features, producing a probability of object identification corresponding to the tasks of OP. Finally, Layer G refers to data visualisation, which for an AI-enabled vision system involves presenting the results of information analysis for OP.

5.2. Addressing privacy concerns for human-centered order picking (RQ2)

Regarding RQ2, the findings indicate that addressing privacy concerns is indispensable and must comply with existing legal frameworks. Recent advances in hardware, computational calculations, and ML underscore the importance of AI-enabled vision systems for generating and acquiring information (Penumuru, Muthuswamy, and Karumbu 2020). However, privacy concerns, including those undermining the rights of staff to data protection and privacy, constitute a major barrier for the deployment of these systems (Chatterjee, Krit, and Raut 2022). These concerns include the legal and privacy implications associated with evaluating and predicting of staff performance based on individual behaviours and traits (European Parliament, Council of the European Union 2024). In agreement with De Lombaert, Rijal, et al. (2024), this study argues that the correct application of data protection principles in AI-enabled vision systems can simultaneously enhance operational efficiency and support staff well-being in warehouses.

Empirical results indicate that AI-enabled vision systems, when applied to OP, fall under the scope of the recent AI Act. These systems are characterised by their autonomous and adaptive operation, as well as their ability to process inputs to generate predictions, recommendations, or decisions that influence staff behaviour (European Parliament, Council of the European Union 2024). These systems are classified as high-risk systems that affect work-related relationships based on data collected from staff.

This study proposed four activities addressing privacy concerns of AI-enabled vision systems in OP. These activities leverage the recommendations of the EDPS concerning data protection and specify measures for the layers (A to G) a human-centered architecture proposed by Rožanec et al. (2023). The first activity concerns informing staff about the purpose of generating and analyzing information with AI-enabled vision systems. This should describe the objective of the analysis, its expected duration, and procedures. It requires voluntary consent with the right of staff to decline or withdraw from participating in the analysis without penalty. This is essential for ensuring compliance with general data protection regulation (GDPR), safeguarding staff from invasive monitoring practices, and clearly defining the means and purposes of data processing (De Lombaert, Rijal, et al. 2024). Additionally, it requires explicit labelling of areas of a warehousing using AI-enabled vision systems. This activity corresponds to layer A involving the physical elements a warehouse.

The second activity involves limiting data acquisition by AI-enabled vision systems. The literature refers to these activities by the term redaction and involves the activities preserving staff privacy following their consent to use these systems (Gupta and Prabhat 2023). Redaction may include restricting the detection area of vision systems and applying computational obfuscation to datasets in cameras, video processors, databases, or user interfaces. This study suggested implementing redaction at layer B (data acquisition), as addressing privacy close to the data source likely reduces personally identifiable information at later stages of the architecture. This is important for minimising the collection of personal data beyond the duration strictly necessary for fulfilling the intended purpose of data processing (De Lombaert, Rijal, et al. 2024). The results showed that redaction may introduce computational costs, but alternatives such as limiting AI-enabled vision systems to specific time periods can mitigate these costs.

The third activity involves anonymizing data generated by AI-enabled vision systems with anonymous

datasets. The study proposed that these systems generate and analyze data that describe visual information without capturing specific content. Extending prior research on anonymous datasets in manufacturing (Castro, Cardoso, and Andrade 2021), the study applied this approach to OP contexts. The results demonstrated that anonymization can be achieved by generating and analyzing geometrical positional data of human key points in a time series. This approach addresses concerns over AI-enabled vision systems, specifically the risk of using datasets that could enable individual identification for scrutiny, micro-management, or punitive measures. This activity is critical for enabling systemic data collection, ensuring that data remains untraceable to individual workers while increasing the visibility of warehouse processes and operational features (De Lombaert, Rijal, et al. 2024). This activity corresponds to Layer C that focuses on data management.

The fourth activity addresses server hosting and storage for data generated by AI-enabled vision systems. This study proposed using local servers to address two concerns identified in the literature. These concerns include secure storage of data from visual information and hosting applications that facilitate information analysis. By storing data and applications on a local network, this approach reduces the exposure of sensitive data and improves control over data access, usage, and storage duration. The results indicate that this activity affects layers B (data acquisition), C (data management), E (data processing), F (data analytics), and G (data visualisation and user interface). Figure 8 depicts the proposed framework for generating and analyzing information using AI-enabled vision systems. The framework incorporates the different layers of a human-centered architecture to monitor and address privacy concerns in OP.

5.3. Implications to human factors and outcomes

The findings of the study suggest that AI-enabled vision systems can better consider human factors in OP. This is achieved by capturing real-time data and transforming it into digital information. The results indicated that these systems do not address all human factors in OP equally. These systems generated information by identifying fundamental motion segments in OP. They can directly recognise motions related to working postures such as travelling, carrying, pushing, pulling, stretching, lifting, or bending. Accordingly, these systems provided benefits primarily to the physical aspects of OP. In agreement with Konstantinidis et al. (2023) and Ciccarelli, Papetti, and Germani (2023), the results of the study suggested that information provided by these systems may be insufficient to assess conditions of fatigue, mental load, boredom, or information perception. The findings indicated that AI-enabled vision systems may identify tasks in OP that consistently result in incidents or deviations from desired performance. However, additional sensing devices are required to evaluate cognitive, perceptual, or psychosocial aspects.

Existing literature suggests that AI-enabled vision systems can improve performance in a human-centered approach to manufacturing (Wierschem, Jimenez, and Méndez Mediavilla 2020). These systems are considered capable of self-adaptation during the evaluation of tasks, and they may aid staff in identifying problems and inform them about potential improvements in real-time, such as, comparing the productivity between experienced and novice workers, ensuring safe conditions, or aiding staff achieve faster outcomes (Wierschem, Jimenez, and Méndez Mediavilla 2020). The findings of this study indicated that information generated by AI-enabled vision systems may contribute to distinct outcomes of OP. That is, the activities of image acquisition and pre-processing may be used as inputs to the analysis of outcomes of OP so long as these focus on tasks executed by staff. However, identifying and evaluating outcomes (e.g. performance, quality, or health) automatically requires individual analysis. Achieving this would involve unique activities for segmentation, feature extraction, and classification tailored to each outcome. This is important as significant resources may be necessary for interpreting, learning, and executing the results of AI-enabled vision systems enhancing a human-centered approach.

5.4. Managerial contributions

The findings of this study highlighted three considerations that can aid managers and engineers in planning and preparing for AI-enabled vision systems in OP. These systems enable managers to acquire data that are otherwise unavailable, based on assumptions, or only gathered periodically, potentially resulting in bias. Accordingly, the findings present an alternative to existing practices that are resources consuming, costly, and manual. By exemplifying the use of these systems in a pilot study, the results demonstrated the feasibility of using these systems in warehouses. However, further analysis might be needed for the utilisation of multi-camera system and for ensuring the functionality and maintenance of cameras in different or more complex industrial environments. This description and exemplification of activities to generate and analyze information about OP can help managers and stakeholders understand the implications of using AI-enabled vision systems in larger or smaller applications across different sectors.

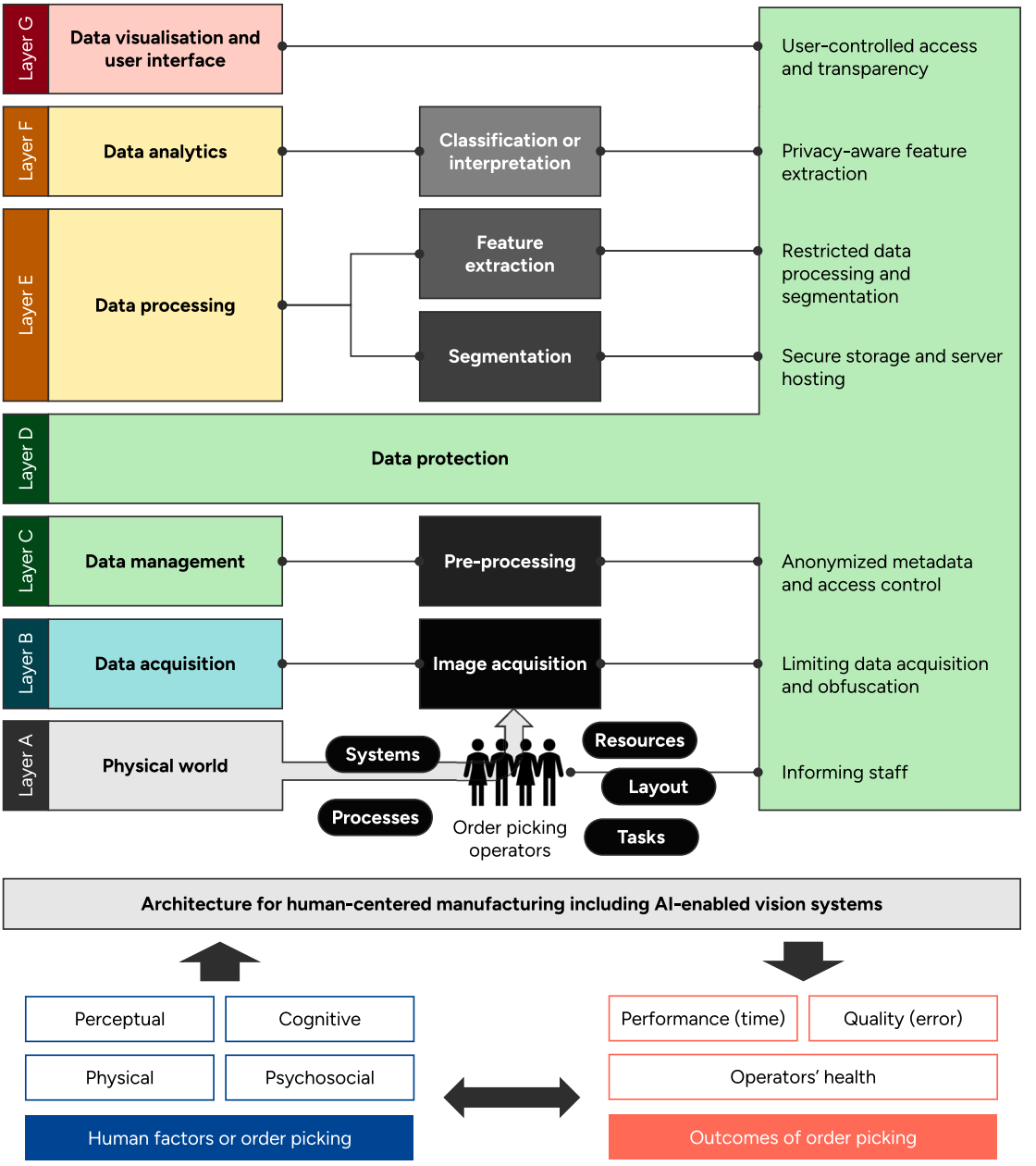


Figure 8. Data generation and analysis framework for AI-enabled vision systems in human-centered order picking within warehouses.

This study developed an architecture for human-centered manufacturing including the use of AI-enabled vision systems for OP. This architecture describes the components, interconnection, interactions and constraints of these systems. Managers can use these findings to evaluate digital capabilities, competence of staff, and resources necessary for implementing AI-enabled vision systems in similar applications as well as other sectors involving human-centered OP. Managers can apply these findings to assess risks, develop basic prototypes, and identify challenges. This may avoid costly changes when projects that incorporate AI-enabled vision systems are implemented.

The study observed that AI-enabled vision systems constitute an important supportive technology for managing the operations of warehouses. Consistent with Sharma, Zhang, and Rai (2021), these findings indicate that manufacturing organisations can gain a competitive edge by implementing AI-based solutions that enhance staff cognitive capabilities. Regardless of a system's size, staff experience, intuition, and knowledge remain essential for identifying OP tasks and training AI-based vision systems to perform this function. When trained and verified, these systems can relieve staff of repetitive tasks, enabling experienced personnel to focus on identifying root causes of issues in OP and engaging

with operational staff to better consider well-being. Managers can benefit from simultaneously developing the technological capabilities required to implement AI-enabled vision systems and the roles and responsibilities of staff managing the operations of warehouses as well as human-centered OP in other applications and sectors. This approach contrasts with current practices by involving managerial, technical, and operational staff as co-creators in the development of AI-enabled vision systems.

5.5. Limitations of the study and future research directions

The findings of the study originate from a pilot study in a laboratory environment. This approach presents three limitations that future research can address. First, the generalizability of the results is limited by the scope of the sample. This constraint originates from the use of DSR as the methodological approach, which was appropriate for an initial exploration of AI-enabled vision systems. However, future studies could expand sample sizes and explore different contexts to test the generalizability of these findings. For instance, subsequent research might examine AI-enabled vision systems in real industrial settings, assessing their scalability and alignment with Industry 5.0 principles and sustainability goals. A logical next step would be to develop a prototype of these systems for testing in a factory environment, which could help evaluate long-term effects on human factors and operational performance outcomes not observable in a controlled lab setting.

Second, the generalizability of the results is also affected by contextual limitations, including the choice of industry. The study focussed on OP in automotive manufacturing, a labour-intensive environment characterised by picker-to-parts practices. Key characteristics include warehouse-to-assembly supply of thousands of parts, kitting, strict time constraints, repetitive travel patterns, and consistent part presentation. To extend its findings, future studies should explore other labour-intensive industries, such as consumer goods, retail, and food production, where product variety and order diversity are more pronounced. Examining these sectors could help determine whether the proposed activities for monitoring tasks and addressing privacy concerns remain effective across distinct contexts.

Third, future studies could address potential sources of bias. For example, while this study focussed on generating and analyzing information in OP, it did not examine how this information affects staff when used as feedback. To explore this dimension, future research could apply alternative methodological approaches–such as participatory research, case studies, action research, or human-centered design–to gain insights from staff feedback. Future studies could investigate the experiences of staff interacting with AI-enabled vision system capabilities (e.g. acquiring, handling, and using information), as well as how these systems account for human factors (e.g. perceptual, cognitive, physical, and psychosocial) and influence OP outcomes (e.g. performance, quality, and worker health).

6. Conclusions

Research increasingly highlights the need to transform practices in warehouses beyond an emphasis on operational goals by prioritising the well-being of staff. The literature highlights the role of digital technologies in enabling this transformation as characterised by the concept of Industry 5.0. However, few studies have explored how information generation and analysis originating from the use of digital technologies change in this paradigm. This study addressed this gap by analyzing the use of AI-enabled vision systems to enhance the generation and analysis of information in OP. It combined an architecture for human-centered manufacturing and existing understanding of the activities of AI-enabled vision systems, and proposed a set of characteristics for generating and analyzing information in OP. The study discussed the findings originating from the development of a pilot study in a laboratory environment by applying a DSR approach that included the collaboration between academic and industrial actors. Additionally, the study proposed four activities applicable to an architecture for human-centered manufacturing for addressing privacy concerns originating from the use of AI-enabled vision systems.

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Data availability statement

The data that support the findings of this study are available from the corresponding author, Erik Flores-García (efs01@kth.se), upon reasonable request.

Disclosure statement

No potential conflict of interest was reported by the author(s).

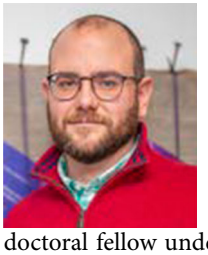
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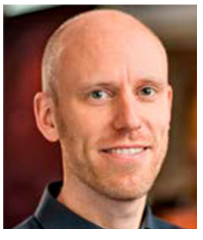
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Appendices

Appendix 1. ANN justification and python code for feature extraction of order picking tasks

ANNS present four advantages when applied to AI-enabled vision systems for task classification (Alam et al. 2020; Mikhaylenko, Kharchenko, and Shcherbinin 2020), which are particularly relevant when monitoring staff movements in OP operations. First, they integrate feature extraction, dimensionality reduction, and classification into a unified process (Verdhan 2021). Target features can be learned directly from input data and classification targets without application-specific feature extractors. Second, ANNs are capable of capturing complex nonlinear relationships between inputs and outputs, enabling interpolation to infer missing signals. Despite these strengths, ANNs face several limitations, as noted by Ansari (2023) and Verdhan (2021) identified several challenges of ANNs. Processing in real-time is computationally expensive, and the requirement for large labelled datasets makes data preparation both costly and time consuming. Moreover, accuracy is affected by environmental factors such as occlusion, lighting, and camera angles. ANNs also lack interpretability, making error analysis and resolution difficult. Finally, monitoring human activities require continuous model updates and retraining, resulting in significant resource requirements. This study applied a Python code to evaluate and visualise the time-series data of the 17 human key points obtained during segmentation. This code is described as follows:

```
# Drop the Serial Number column (Column
    1) and unnecessary features
data = data.iloc[:, 1:]
# Exclude 'Round Time' and 'Round number'
     from features if they exist
columns_to_exclude = ['Round Time', '
    Round number']
data = data.drop(columns=[col for col in
    columns_to_exclude if col in data.
    columns])
# Handle missing values in features (
    Columns 5-15): Impute numeric data
    with mean
features_to_impute = data.iloc[:, 4:14]
    # Features columns
imputer = SimpleImputer(strategy='mean')
imputed_features = pd.DataFrame(imputer.
    fit_transform(features_to_impute),
    columns=features_to_impute.columns)
# Explicitly cast the imputed features
    back to their original data types
for col in imputed_features.columns:
    if np.issubdtype(data[col].dtype, np.
    integer):
        imputed_features[col] =
    imputed_features[col].round().astype(
    int)
    else:
        imputed_features[col] =
    imputed_features[col].astype(float)
# Combine the imputed features back with
    the rest of the dataset
data.iloc[:, 4:14] = imputed_features
# Define the feature set by excluding
    label columns and excluded columns
feature_columns = data_cleaned.columns.
    difference(label_columns)
X = data_cleaned[feature_columns]  #
    Features are all columns except
    labels and excluded columns
```

Appendix 2. Interview protocol

Background

- Describe your educational and work background, especially any experience relevant to warehouse operations.
- Is there anything about your background or upbringing that you think influences how you work or interact in the workplace?
- Are there any health-related considerations you'd like to share that affect how you perform your job tasks?

Order picking tasks

- **Setup** What activities do you perform before you start picking items?
- **Travel** How do you move through the warehouse and handle the items physically (e.g. carrying, pushing, reaching, etc.)?
- **Search** How do you locate and recognise the items you need to pick?
- **Pick** Once you've found an item, how do you retrieve and prepare it for the next step in the process?

Human factors

- In your opinion, what skills or qualities are important for someone to be an effective order picker?
- How do sensory factors like sight, sound, or touch influence your ability to pick orders?
- What are the main challenges you experience during the picking process, and how do you typically handle them?
- What do experienced operators do differently that contributes to a smooth picking process?
- How do you typically learn and adapt to the tasks involved in order picking? What helps you remember processes and perform efficiently?
- Are there parts of the picking process that are physically more difficult or easier for you?
- How do you typically feel physically at the end of your workday?
- Do you often work in positions or postures that are uncomfortable or tiring?
- Can you describe any physical strains or risks you experience during order picking tasks?
- What moments during your workday do you find most satisfying or most difficult?
- Are there aspects of your work that you find particularly stressful or motivating?

Outcomes of order picking

- How do you assess whether you're doing a good job when picking orders?
- Are there specific outcomes you track (e.g. time, accuracy, speed)?
- When errors in picking occur, what do you think contributes to them?
- Are there any physical aspects of your job that help or hinder your ability to work effectively?

Appendix 3.

Quotes of staff at case company

Table A1. Negative working conditions affecting the well-being of staff in order picking at the company.

Negative working conditions	Description
Maintaining pace of picking in a shift	*"Maintaining the pace of picking during the shift and making few mistakes is difficult. Specially when you are new to a warehouse. It takes some time to know where the materials are."*
	"We know that our performance drops after lunch or close to the end of the shift. It is mentally challenging and you are tired. You can make silly mistakes like forgetting to use the lifting tools, or making picking motions that are not approved."
Uneven work pace throughout shift	*"There is a big difference in pace with a slow and high pace in the assembly line. On a high demand we really need to focus. It always take a while to adjust…This also happens with the type of product mix. Some vehicles have more options than others, these means additional picks in the same time. It can get stressful and we need to work fast."*
Location of parts in warehouse	*"The engineers do a good job at designing the locations of the warehouse, but over time the items change place. It can take a while to make adjustments that work for the product, the warehouse, and us [staff]. It takes time to make this analysis and it would be good to do it frequently."*
Excessive walking to locate parts	*"We do not have long aisles in the warehouse. The area is only a couple of meters long. However, we do need to walk back and forth a lot. It is a hard floor and you stand up all day. You do get tired and it takes sometime to know the location of parts."*
Validation of quantity and quality of parts	*"Controlling quality, quantity, and part number is something we do all the time and after every pick…This is very time consuming. It is also quite challenging. Our products have modular designs and the components we pick might look exactly the same on the outside but their function or performance is different. You really need to focus on part numbers to avoid making mistakes…Validating parts requires a lot of time both while we pick and after we are done with an order but it is necessary."*

Table A2. Problems when generating and analyzing information and privacy concerns.

Identified problem	Description
Limited visibility	*"We have different picking methods including sequence picking, kit picking, and dynamic sequencing. While we design our warehouses in alignment with these methods having insight about the actual tasks performed by staff is a current challenge that we have not managed to solve. We know however that digitalisation is key to increasing visibility of OP."* – Logistics developer
	"We have managed to establish a way of working that we replicate in all of our warehouses for OP. But we cannot tell if this is really the best way to do things. We have very high ambitions for both staff well-being and pace of production and these can be conflicting goals. For example, lifting tools are heavy, slow, and big but we want them in our factory. Better data that can help us achieve both goals." – Logistics manager
	"We currently record our staff with hand held cameras. This helps us define the pace of OP and we try to hold a standard… It is one thing to record in a perfect setting and a very different thing doing so at a regular pace of production. When we make changes: is it really better than the previous way of working? Does it balance the work of staff? We like to think so, but we cannot verify this as we do not have data." – Time-data management engineer
Analysis of tasks in OP	*"Recording is extremely time consuming. One minute of video takes 30 min of analysis. We need to look at the video and determine what our staff is doing. It is very common that people have different ways of executing the same task and the analysis can take a very long time."* – Time-data management engineer
Time and effort for detecting problems	*"It is only a handful of specialists who can do it in the entire site. Some times we book a recording, have everything in place, and then… nothing works as planned in OP. It is a huge investment of time for everyone…We are overloaded with analysis work."* – Digitalization manager
Privacy concerns	*"We always ask staff in the warehouse and managers about recording. We have a procedure. It can happen that staff do not agree to a recording and that is ok. It is part of the procedure. The results of our recording always need to be discussed with staff and their managers. We need to comply with GDPR and maintain privacy of the staff. We do not want to single out the low performance of an individual. Instead, we aim to provide changes to OP that provide a collective benefit for a team."* – Time-data management engineer

Appendix 4. Example of results of the classification of tasks

The following table presents empirical data on the classification of OP tasks and their corresponding times in a laboratory environment. The table includes five picking lists (A to E), each specifying the part number, zone, location, and crane usage for each part. Additionally, the table provides the time (in seconds) for search, pick, travel, and setup tasks, along with the total time. These values represent the averages from ten rounds of OP for each picking list, which were randomly assigned to ten participants in the study. Figures A1 and A2 compare the distribution of OP task times from the pilot study with those reported by Tompkins (1996), expressed as percentages. The findings suggest a similar distribution of time across search, pick, travel, and setup tasks in both studies. However, the pilot study indicates a lower proportion of time spent on travel, and a higher proportion spent on search and setup tasks. This study did not investigate the underlying causes of these differences; further research is needed to explore them. One possible explanation is

Table A3. Parts and time of tasks in order picking of pilot study.

Picking list	Part number	Zone	Location	Search time (s)	Pickup time (s)	Travel time (s)	Setup time (s)	Total time (s)	Crane
A	772078	2	Top shelf	4,44	2,12	8,49	4,25	19,30	0
A	791565	3	Bottom shelf	5,50	2,63	10,52	5,26	23,90	0
A	610539	2	Bottom shelf	2,48	1,19	4,75	2,38	10,80	0
A	695625	1	Top shelf	3,75	1,79	7,17	3,59	16,30	0
A	520803	8	Bottom shelf	2,21	1,06	4,23	2,12	9,62	0
A	653209	8	Bottom shelf	2,62	1,25	5,02	2,51	11,40	0
B	310493	3	Top shelf	3,63	1,74	6,95	3,48	15,80	0
B	791565	3	Bottom shelf	2,25	1,07	4,30	2,15	9,77	1
B	536893	10	Top shelf	1,35	0,65	2,59	1,29	5,88	0
B	239871	6	Top shelf	2,10	1,00	4,01	2,00	9,11	0
B	701368	10	Bottom shelf	1,81	0,87	3,47	1,73	7,88	0
B	653209	5	Bottom shelf	1,12	0,54	2,15	1,07	4,88	0
B	157250	5	Top shelf	1,48	0,71	2,83	1,42	6,44	0
B	418943	11	Bottom shelf	2,17	1,04	4,15	2,08	9,44	0
C	772078	2	Top shelf	1,44	1,69	2,75	1,38	7,25	0
C	985524	4	Top shelf	1,18	1,56	2,25	1,13	6,12	0
C	536893	10	Top shelf	1,87	0,89	3,57	1,79	8,12	0
C	520803	8	Bottom shelf	2,62	1,25	5,02	2,51	11,40	0
C	520416	7	Top shelf	1,32	1,63	2,53	1,27	6,75	0
C	420381	11	Top shelf	2,00	0,96	3,85	1,93	8,75	0
D	297602	9	Top shelf	5,41	2,59	10,34	5,17	23,50	0
D	310493	3	Top shelf	9,69	4,63	18,53	9,26	42,11	0
D	948071	9	Bottom shelf	7,45	3,56	14,26	7,13	32,40	0
D	791565	3	Bottom shelf	2,88	1,38	5,50	2,75	12,50	1
D	714035	6	Bottom shelf	2,16	1,04	4,14	2,07	9,41	0
D	520803	8	Bottom shelf	2,15	1,03	4,12	2,06	9,36	0
D	420381	8	Top shelf	2,71	1,30	5,19	2,60	11,80	0
E	520803	8	Bottom shelf	1,92	0,92	3,67	1,83	8,34	0
E	948071	9	Bottom shelf	7,20	3,44	13,77	6,89	31,30	0
E	536893	10	Top shelf	4,00	1,91	7,66	3,83	17,40	0
E	420381	11	Top shelf	1,67	0,80	3,20	1,60	7,28	0
E	610539	2	Bottom shelf	4,09	1,96	7,83	3,92	17,80	0
E	157250	5	Top shelf	2,32	1,11	4,43	2,22	10,07	1
E	239871	6	Top shelf	2,69	1,29	5,15	2,57	11,70	0

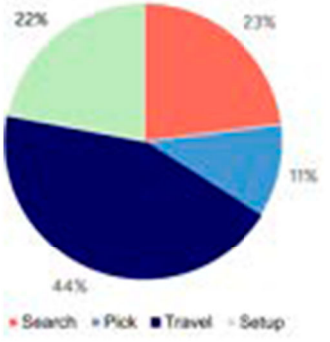


Figure A1. Time of order picking tasks in pilot study.

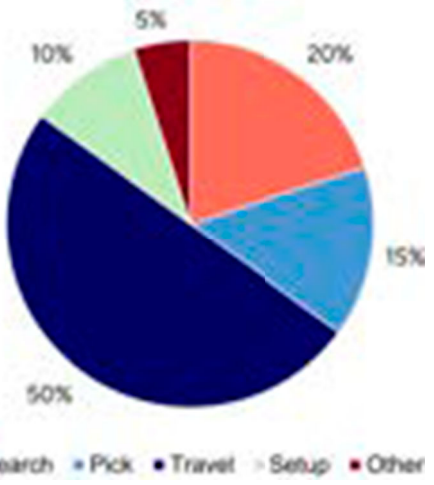


Figure A2. Time of order picking tasks according to Tompkins (1996).

that the pilot study was conducted in a laboratory setting unfamiliar to participants. As a result, participants required more time to locate and set up materials, while the smaller scale of the laboratory, compared to real-life warehouses, reduced travel time.